

REACTION RATE

- Which one of the following increases the rate of reaction, does not take part in reaction products, and does not undergo any change during the reaction ?
A) electrolytes B) catalysts C) solvents
D) nonelectrolytes E) none of them
- The mixture of hydrogen and oxygen in glass tubes produces water at 300 °C after a few days. If temperature is increased to 500 °C, water forms after a few hours. Which factor increases rate of the reaction?
A) the concentration of reactants B) temperature
C) the interacting area of reactants D) catalyst
E) the nature of reactants
- Which one of the following is the formula of rate ?
A) $V_{t_2} = V_{t_1} \cdot \gamma^{(t_2 - t_1)/10}$ B) $V = m/p$ C) $M = m/V$
D) $V = \Delta C/\Delta t$ E) none of them
- In two different test tubes, there is powdered zinc. 20 % sulphuric acid is added to the one of them on the other hand, 10 % sulphuric acid is added to another one. What can be said about the rate of reaction ?
A) rates differ according to the temperature.
B) rates differ according to the concentration of reactants.
C) rates differ according to the catalysts.
D) rates differ according to the the nature of reactants.
E) rates differ according to the interacting area of reactants.
- Given the reaction ;
$$2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$$

The concentration of NO decreases from 0,8 mol/L to 0,2 mol/ L in 20 sec. What is the average rate of the reaction in mol/L.s?
A) 0,04 B) 0,032 C) 0,03 D) 0,02 E) 0,01
- If temperature increases 10 °C, rate increases two times. How many times does rate increase if temperature is increased from 20 °C to 40 °C ?
A) 2 B) 200 C) 3 D) 4 E) 100

7. What is the unit of rate ?

- A) mol/L B) g/100 g H₂O C) mol/L.s
D) J E) J/mol

8. Given the reaction ; $\text{N}_2 + \text{O}_2 \rightarrow 2\text{NO}$

I. The concentration of nitrogen increases two times.

II. The concentration of nitrogen increases two times and the concentration of oxygen decreases two times.

How does rate change ?

- | I | II |
|------------------------|----------------------|
| A) increases two times | increases four times |
| B) decreases two times | no change |
| C) increases two times | decreases two times |
| D) no change | increases two times |
| E) increases two times | no change |

9. Given the reaction ; $\text{A} + 2\text{B} \rightarrow \text{C}$

The initial concentration of B is 6 mol/L. After the beginning of the reaction in 10 seconds the concentration of C becomes 2 mol/L. What is the average rate of reaction ?

- A) 0,6 B) 2 C) 0,3 D) 0,2 E) 3

10. One test tube contains powdered iron another test tube contains iron chunk. Then, hydrochloric acid is added to each test tube. What can be said about the rate of reaction ?

- A) rates differ according to the temperature.
B) rates differ according to the concentration of reactants.
C) rates differ according to the catalysts.
D) rates differ according to the the nature of reactants.
E) rates differ according to the interacting area of reactants.

11. If temperature is increased by 10 °C, rate increases four times. How much temperature must be increased so as to increase rate 16 times ?

- A) 40 °C must be increased
B) 4 times must be increased
C) 16 times must be increased
D) 30 °C must be increased
E) 20 °C must be increased.